

ALEJANDRO DIAZ

U.S. Citizen | Active Secret Security Clearance (DoD)

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EDUCATION

University of Texas at Austin

Bachelor of Science in Mechanical Engineering

Track: Quantitative Systems Engineering & Management | Minor: Entrepreneurship

Dec 2026

GPA: 3.86/4.0

WORK EXPERIENCE

Saronic Technologies | *Manufacturing Engineering Intern*

Jun 2026 – Aug 2026

- Owned an RF-antenna torque tool from first-principles design analysis through in-house sourcing, eBOM, drawings, and an ECO procedure-change proposal; eliminated antenna marring (0 cases since rollout) and reclaimed ~49 hr/yr fleet-wide
- Led a DMAIC-framed cost study of a \$1.28M/yr SciGrip structural-adhesive program, building station-level consumption models that identified \$923K/yr in achievable material savings (47-day payback, 677% ROI) while correcting a \$143K overstatement by isolating achievable from theoretical coverage
- Architected and shipped 7+ Palantir Foundry and full-stack tools (React/TypeScript, Node/Express, Python), including custom ontology objects, pipelines, and Workshop dashboards adopted by active users across Engineering, PMO, and Manufacturing Engineering; replaced manual ERP lookups with sub-second (0.2s) part search across 79,549 PO records and 15 product lines
- Built a first-principles bearing-stress model gating a fleet-wide fastener DFMA initiative; screened 7 joint types and baselined 218 fastener part numbers, identifying ~\$13.3K/yr in savings and 167 washers eliminated per vessel by converting nut/washer stacks to flange hardware

Northrop Grumman Corporation | *Continuous Optimization & Refinement Engineering Intern*

May 2025 – Jul 2025

- Developed a Python multivariate regression model integrating SMR and laser gauge metrology data for F-35 fixture loading vector estimation and alignment process optimization; feasibility study validated a novel corrective action methodology targeting an NC condition affecting ~33% of units, each carrying 2–10 hr of unplanned rework
- Automated quality management workflows using Power Automate, decreasing QA response times by 21.1% within three weeks through standardized cross-functional digital reporting procedures
- Contributed to Graph-RAG LLM architecture for F-35, B-2, and E-2D programs; defined prompt hierarchy and validation protocols for procedure-driven retrieval and response generation in classified manufacturing workflows
- Developed Power BI executive dashboards with automated data pipelines to surface production KPIs, track manufacturing defects, and automate a manual reporting process saving ~312 hr annually

Northrop Grumman Corporation | *Manufacturing Engineering Intern*

May 2024 – Jul 2024

- Designed DFM-optimized 3D-printed production aids with integrated precision depth gauges, reducing tool validation time by 25%; directed work-cycle studies and standardized production processes for F-35B tooling
- Conducted root cause analysis on BusinessObjects manufacturing data, identifying sealant disbond failure modes and quantifying \$800K+ annual rework and 2,800 labor hours; developed Tableau dashboards correlating CMM Romer Arm metrology trends, implementing corrective actions reducing unit rework by 10–20 hr

Agave Worldwide LLC | *Lead Product Development Engineer*

Jul 2024

- Applied DFMA to design a 36" smokeless fire pit in SOLIDWORKS; developed parametric sheet metal models with integrated bending and die design, resolving airflow distribution and thermal management for high-rate production

RELEVANT EXPERIENCE

Texas Rocket Engineering Lab | *Lead Mechanical Design Engineer*

Jan 2025 – Present

- Sole design engineer for the 3DOF cold gas RCS; derived mechanical state equations for required thrust, pressure, and mass flow to enable software-team PID attitude/roll stabilization
- Led structural redesign of the TVC actuator subsystem for the Halcyon MK1 liquid bi-propellant rocket; executed multi-criteria trade studies on material properties and load margins to determine lightweight actuator architecture
- Programmed parametric MATLAB propulsion models to optimize nitrogen tank sizing and performance envelope

Longhorn Racing Electric | *Dynamics Mechanical Design Engineer*

Sep 2023 – Feb 2025

- Redesigned front and rear unsprung assemblies via ANSYS FEA and GD&T-driven optimization, achieving 44% rear assembly and 36% front upright weight reductions under fatigue, thermal, and dynamic load requirements
- Developed magnetic flux analytical frameworks to redesign wheel speed sensor architecture, achieving $2.5\text{--}5\times$ signal amplitude improvement over prior design; selected catalog magnets and validated on vehicle

LEADERSHIP & SERVICE EXPERIENCE

Beta Theta Pi Fraternity | *Vice President of External Programming*

Nov 2023 – Present

- Directed planning and operations for a 250-person charity event; managed a \$30K budget, negotiated contracts for 10%+ cost savings, and raised \$35K for the Texas 4000 For Cancer organization

ADDITIONAL

CAD & Analysis: CATIA, NX, SOLIDWORKS, AutoCAD, Fusion360, OnShape, ANSYS, PolyWorks, GD&T

Software & Tools: Python, MATLAB, Java, Palantir Foundry (AIP), Power Automate, Power BI, Tableau, BusinessObjects,

Teamcenter, Arena PLM, Excel, GitHub, Jira, Spanish (conversational)

Manufacturing: Lathe, CNC Mill, CNC Router, Waterjet, Laser Cutting, FDM 3D Printing, Romer Arm, SMR Tracker

Honors: FRC 6800 Valor: Engineering Excellence, World Championship (2023); UIL Robotics Gold (2022)